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Statistics and Probability : Study Notes

1) Introduction to Statistics

Statistics is the science of collecting, summarizing, analyzing and interpreting data so we can make decisions.

working flow : Collect data, summarize it, analyze it, make conclusion.

Why it matters in Machine learning:

- understand the data (averages, spread, patterns)
- Handles uncertainty (probabilities)
- Find relationships between variables
- Make better predictions
- Test whether one model is better than another.
- Support real-world decision)

Two branches:

- Descriptive statistics summarizes and describe the data you already have (median, spread, charts)
- Inferential statistics uses a sample to draw conclusions about a larger population.

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Topics ~~cont~~ covered under descriptive statistics: central tendency, dispersion, quartiles and percentiles, outliers and skewness, frequency distribution, graphs and charts.

Measures of central tendency

Central tendency is the value around which the data tends to cluster, the typical or central value

Mean ($\bar{x}$)

- Defination: The sum of all values divided by the number of values.

$$\bar{X} = \frac{\Sigma x}{n} \quad \text{(Normal)}$$

$$\bar{X} = \frac{\Sigma (x-f)}{\Sigma f} \quad \text{(frequency table)}$$

Eg: marks of 20 students gave ~~Σf~~

$\Sigma fx = 1520$ so mean($\bar{x}$) $= \frac{1520}{20} = 76$

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Application

Data summarizing, filling mission values, standardization (feature scaling), comparing groups and outlier detection

weakness

The mean is sensitive to outliers.

Median

The middle value of sorted data.

- Odd n : The value at position (n+1)/2
- Even n : The average of the value at positions n/2 and (n/2)+1.

Eg:

10, 20, 30, 40, 50, 60 has median,

$$M_d = \frac{30 + 40}{2} = 35$$

Strength:

The median is robust to outliers

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Mean vs Median (Salary example)

For six famous people's salaries (24000, 260, 135, 120, 75, 50 million USD) the mean is about $107 million but the median is 97.5 million. One extreme value ruins the mean, while the median stay sensible.

Cumulative Frequency (cf)

The running total of frequencies. It tells how many observations are less than or equal to a value

Each cf = current frequency + all previous frequency

- It is used to locate the median and quartiles in a frequency table.

Mode

The value with the highest frequency.

Eg.

In the marks data, 85 occurs 6 times

So,

Mode = 85

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Uses:

Most common category or choice of the users. (popularity)

9. Measures of Dispersion (Spread)

Dispersion tells how spread out the data is around the centre.

Range

Range : $Max^m - Min^m$

Example:

min 95, max 80, range = 35

Larger range means more spread, uses only two values so, outliers affects it.

Variance

The average squared distance of values from the mean.

formula:

$$\sigma^2 = \frac{\Sigma (x - \bar{x})^2}{N}$$

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Example:

Data 48, 49, 50, 51, 52 has.

$\bar{x} = 50$

Square Difference = 4, 1, 0, 1, 4,

which,

sum to 10, so variance = $\frac{10}{5}$

$= 2$

small variance means consistent data. Large variance means high ~~variable~~ variability.

Problem: The unit is square, which is hard to interpret.

Standard Deviation (SD)

The square root of the variance, the typical distance of values from the mean.

$$\sigma = \sqrt{\text{variance}}$$

Eg:

$\sqrt{2} = 0.6$

$\therefore \sigma = 1.41$

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It is in the same unit as the data, so it is easier to interpet.

Not every value must lie within 1 SD of the mean.

Interquartile Range (IQR)

The spread of the middle 50% of the data, IQR = $Q_3 - Q_1$ (see next section)

Percentiles and Quartiles

Percentile

The pth percentile is the value ~~between~~ below which p% of the data lies. It shows the position of a value in sorted data.

Ex:

In the sorted marks 40, 45, 50, 55, 60, 65, 70, 75, 80, 90 the 70th percentile is 70 meaning 70% of value ~~as~~ are at or below 70.

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Quartiles

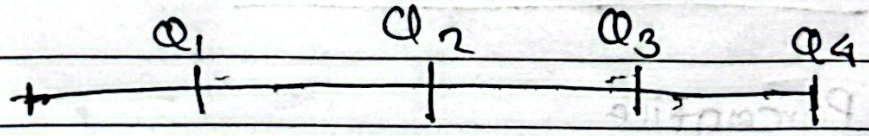
Quartiles divide sorted data into four equal parts. with n observation.

Q_1 = 25th percentile at $(n+1)/4$

Q_2 = 50th " " $(n+1)/2$
↳ (median)

Q_3 = 75th " " $3(n+1)/4$

Q_4 = 100th pe " " the maximum value.



Ey:

2, 4, 5, 8, 10, 12

$n = 7$

$Q_1 = 4$

$Q_2 = 8$

$Q_3 = 12$

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Outlier Detection

An outlier is a value unousually far from the rest of the data.

IQR method

$IQR = Q_3 - Q_1$

Lower bond $= Q_1 - 1.5 \times IQR$

Upper bond $= Q_3 + 1.5 \times IQR$

Any value below the lower bound or above the upper bound is potential outlier

Z-score method

Flag a value as a possible outlier.
if $|z| > 3$

Score 20 with mean 66.1 and SD is 13.9 gives $z = -3.3$ so, it is a possible outlier.

Donot Delet the outlier immediately. First check wheather it is a data error or a genuine rare observation.

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Skewness

⊗ It describes the asymmetry of distribution. ~~Remember~~ Remember it by the tail the skew has the same name as the side the tail is on.

- Symmetric: both sides are mirror image. Mean = Median = Mode

- Left-skewed: ~~more~~ most ~~values~~ values ~~are~~ high a few very small values form a long tail on the left.

  ⇝ $\bar{X} < M_d$

- Right-skewed: most values are low, a ~~few~~ very ~~fast~~ large value form a long tail on the right.

  ⇝ $\bar{X} > M_d$.

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Covariance

Covariance shows wheather two variables move together in the same direction or in opposite direction:

$$Cov(X, Y) = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{(n-1)}$$

Interpretation

- $Cov > 0$: positive both tend to inorease ~~togeth~~ together
- $Cov < 0$: negative, one increase while other decrease.
- $Cov \approx 0$: no clear linear relationship.

Limitations.

It isnot standardized, depends on the units, so it's size alone docsnot show how strong the relationship is. Example: 1.9 and 1.6 are both +ve and only slightly different.

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correlation (Pearson r)

It is measure the strength and direction of the linear relationship between two quantitative variable.

$$r = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\sqrt{\Sigma (x-\bar{x})^2 \cdot \Sigma (y-\bar{y})^2}}$$

which equals:

$$Cov(X,Y) / (\sigma X \cdot \sigma Y)$$

Key points:

- r range from -1 to +1 and has no unit
- +1 is a ~~perfect~~ perfect +ve linear relation ship, -1 is a -ve one. 0 is no linear relationship.
- r close to +1 is strong positive a small positive r is weak positive, a small -ve r is weak negative, r close to -1 is strong negative.

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Example:

$\Sigma(x-\bar{x})(y-\bar{y}) = 6$
$\Sigma(x-\bar{x})^2 = 10$ and,

$\Sigma(y-\bar{y})^2 = 6$

So,

$r = \frac{6}{\sqrt{(10 \times 6)}}$

$= \frac{6}{7.746}$

$= 0.775$, a fairly strong +ve linear relation ship.

9) Z-score, Standardization and Normalization

~~How~~

- Z-score

How many standard deviations a value is above or below the mean.

$Z = (x - \mu) / \sigma$

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$z > 0$ means above the $\bar{x}$, $z < 0$ below $\bar{x}$, $z = 0$ exactly at the mean.

Large $|z|$ means farther from mean.

- **Standardization**

Converts each value to a z-score giving mean 0 and standard deviation 1.

Application: A feature with large numbers (income in $) would otherwise dominate a feature with small number (age in yrs)

- **Normalization** (min-max scaling)

Rescales value to a fixed range, usually 0 to 1.

$$x' = (x - \min)/(\max - \min)$$

e.g. age 25 with min 18 and max = 35,

$$x' = \frac{(25-18)}{17} = 0.41.$$

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When to use which

- Normalization : when you need fixed range such as 0 to 1. Good for image pixels, some ~~neural~~ neural network inputs, KNN and k-means.

- Standardization : when features have different units or scales, ~~or~~ or you want a mean 0 and SD = 1

- Tree-based models (Decision Tree, Random Forest, XG Boost) usually need no scaling.

10) Normal Distribution

A symmetric, bell-shaped, continuous probability distribution where most values are near the means and extremes are rare. It is written $X \sim N(\mu, \sigma^2)$.

Properties

- Smooth, bell-shaped ~~curve~~ curve, symmetric around the mean

- mean = median = mode
- Total area under the curve = 1; 50% of the total data lies on each side of the mean
- Fewer values appear as you move ~~off~~ away ~~the~~ form the mean
- completely defined by two parameters: ~~mu~~ ~~sd~~ μ control s the center and σ σ controls the spread.

$$f(x) = (1/(\sigma\sqrt{(2\pi)})) \cdot e^{\wedge}(-(x-\mu)^2/2$$

Empirical rule (68-95-99.7):

- About 68% of value lies within $\mu \pm 1\sigma$
- " 95% " " " " $\mu \pm 2\sigma$
- " 99.7% " " " " $\mu \pm 3\sigma$

- values beyond $\mu \pm 3\sigma$ are very rare an may be possible outliers.

Real-life example: adult height, exam ~~sc~~ scores in a ~~larg~~ larger class IQ.

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Central Limit Theorem (CLT):

If you take many random samples and compute each sample mean, the distribution of those sample means is approximately normal, even if the original data is skewed or uniform.

1) Probability Basic

Probability measures how likely an event is to happen.

Formula: $P(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$

Range: $0 \leq P(A) \leq 1$

- 0 = impossible (rolling a 7 on a normal die)
- 0.5 = equally likely (heads on fair coin)
- 1 = certain (rolling a number from 1 to 6)

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Example:

- P(even on a die) = 3/6 = 0.5
- P(Ace from 52 cards) = $\frac{4}{52}$

$= \frac{1}{13}$

$= 7.69\%$.

(2) Rule of Probability

1) Range Rule: $0 \leq P(A) \leq 1$
2) Complement Rule: $P(A') = 1 - P(A)$
3) Addition rule (A or B) = $P(A \cup B)$

$= P(A) + P(B) - P(A \cap B)$

4) If A and B are mutually exclusive (cannot happen together),

$P(A \cap B) = 0$

$P(A \cup B) = P(A) + P(B)$

Example: P(roll 1 or 6) = $\frac{1}{6} + \frac{1}{6}$

$= \frac{1}{3}$

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1) Multiplication rule (A and B)

$$P(A \cap B) = P(A) \cdot P(B|A)$$

- If A and B independent,

$$P(A \cap B) = P(A) \cdot P(B)$$

- In ~~if~~ Dependent example:

~~Two~~ Two sixes in a row = $1/6 * 1/6$

$$= \frac{1}{36}$$

- Dependent example:

Two aces in a row without ~~replacement~~ replacement = $\frac{4}{52} \times \frac{3}{31}$

$$= \frac{1}{221}$$

After one Ace is drawn, only 3 Aces remains in 51 cards.

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5) Conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

6) Total Probability:

$$P(A) + P(A') = 1$$

13) Bayes's Theorem

Bayes' theorem updates the ~~pro~~ probability of a th hypothesis after seeing new ~~ed~~ evidence.

$$P(A|B) = P(B|A) \cdot P(A) / P(B)$$

++ It comes from $P(A \cap B) = P(A|B) \cdot P(B)$
$= P(B|A) \cdot P(A)$

Law of total probability,

$$P(B) = P(B|A1) \cdot P(A1) + P(B|A2) \cdot P(A$$

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Example 1:

Two Bags, Bag 1 has 3 red and 1 blue, Bag 2 has 1 red and 3 blue. A bag is chosen at random (prior 50% each).

$$P(\text{Red}) = (3/4)(1/2) + (1/4)(1/2)$$
$$= 1/2$$

$$P(\text{Bag1} \mid \text{Red}) = (3/4)(1/2)/(1/2)$$
$$= 3/4$$
$$= 75\%.$$

19) Quick Formula sheet:

- Mean $= \Sigma x / n$
- Median position $= (n+1)/2$
- Range $= \text{Max}^m - \text{Min}^m$
- Variance $= \Sigma (x - \text{mean})^2 / N$
- SD $= \sqrt{\text{variance}}$
- Quartile position $= (n+1)/4, (n+1)/2, \frac{3(n+1)}{4}$
- IQR $= Q_3 - Q_1$
- $\text{cov}(X, Y) = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{(n-1)}$

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- $r = \frac{cov(X, Y)}{(\sigma x \cdot \sigma y)}$

- $z = \frac{(x - \mu)}{\sigma}$

- min-max = $(x - min)(max - min)$
- Empirical rule = 68% | 95% | 99.7%.
- P(A) = favorable / total
- P(A') = 1 - P(A)
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cap B) = P(A) \cdot P(B|A)$
- $P(A|B) = P(A \cap B) / P(B)$
- Bayes : $P(A|B) = P(B|A) \cdot P(A) / P($